

Access Model of Web Users Based on Multi-chains Hidden Markov Models

Kangfeng Zheng, Yixian Yang
Key Laboratory of network and information attack &
defence technology of MOE ;
Beijing University of Posts and Telecommunications
Beijing, China
kfzheng@bupt.edu.cn
Xiujuan Wang
College of Computer Sciences

Beijing University of Technology
Beijing, China
xjwang@bjut.edu.cn
Kangfeng Zheng, Shize Guo
No. 54 Research Institute of the General Staff of the
PLA ,
Beijing China
kfzheng@bupt.edu.cn

Abstract—How to distinguish abnormal access from normal ones is the key problem in Distribution Denial of service(DDoS) attack detection. This research aims at finding out the major difference between the abnormal access and the normal ones in application layer. To this end , the paper conducts modeling on Web users' access behaviors based on hidden Markov model(HMM) with multiple chains and puts forward a method to identify abnormal access. Tests on simulation data indicate that this model can undo Web users access to a certain degree and discern the abnormal access well.

Keywords- HMM;DoS;Web user access

I. INTRODUCTION

Distributed Denial of Service (DDoS) attack [1] is a common way to attack Internet, and it has become the major threat to Internet security. Researchers have proposed many methods to detect and defense DDoS attack, wherein, references [2-5] study anomaly detection by statistical characteristics of the network traffic; [6,7] realize detection based on the packet feature matching. These detection methods aim at the network layer. However, the application layer DDoS attack realizes attack effect through legal network links and requests instead of depending on imperfect low-level protocols or operating system[8], in this case simply matching the network layer features or traffic statistics cannot detect abnormalities. Noted that the abnormal user access behaviors are different from the normal ones after all, this makes it possible to distinguish the abnormal user access behaviors from the normal ones through conducting modeling on the normal users access behaviors. [8,9] give anomaly detection at application layer. However, similar studies is seldom and references adopt hidden semi-Markov model to model on user access, without considering the Web service content and its influence on user access. Hence existing models are not perfect.

This paper analyzes the characteristics of normal Web users' access behaviors, conducts modeling on Web users' access behaviors based on hidden Markov model with multiple chains and puts forward a method to identify

abnormal access. Besides, the paper trains and tests the model with simulation data. Finally it analyses the model test results.

The structure is as followed: section 2nd gives normal Web users access behavior analysis. Section 3rd establishes the multi-chains HMM model. Section 4th analyzes the tests results and performance. Section 5th gives the conclusion.

II. USER ACCESS BEHAVIOR ANALYSIS

Websites which provide Web services classify pages according to the content in organization. Users' interests often coincide with these categories, namely, the user access behaviors tend to focus on pages of a few classes. User's access process to web pages of each class is in accordance with HMM model. When conducting HMM model, existing DDoS detection methods assume that users' access behaviors agree with the property of Markov chain, i.e., the current state is only relevant to the last one state before it, and the access path can be described by a chain of browsing states in which one state represents a (or multiple) users request(s). But in fact, due to the prevalence of browser with multiple labels, some users tend to browse web pages after opening them at almost the same time which makes the traditional single chain HMM detection model to give error results (observation sequence cannot describe users' real access attempt).

Given a possible example, a user opens one page A, e.g. the homepage of Sohu at moment t_0 and he finds several links to browse, such as cultural link E_1 , real estate link H_1 and IT link I_1 . After that he opens these links separately in three new labels of the browser which correspond to the behavior that E_1 , H_1 , I_1 are clicked at t_1 , t_2 , t_3 respectively and the corresponding states are noted as SE_1 , SH_1 , SI_1 . Then the user starts browsing page E_1 opened just now where he finds another interested page E_2

and clicks it at t_4 (corresponding state is noted as SE_2). Before the content of E_2 is delivered completely the user turns to browse H_1 . Similarly new interested link H_2 might be found in H_1 and clicked at t_5 which corresponds to state SH_2 , and so on. The browsing sequence in classified web pages is shown in Fig.1

In the original model, the resulted state transition sequence and observation sequence of HMM are shown in Fig.2, wherein $\Delta t_i = t_{i+1} - t_i$.

In above Markov chain, the probability of a state SE_2 depends only on its last state SI_1 . Actually, page E_2 is obtained in E_1 which means that the probability of a state SE_2 should depend on SE_1 . Likewise, state SH_2 should rely on SH_1 instead of the named last state SE_2 defined in original model. Therefore,

$$P(S_{i+1} = SE_2 | S_i = SI_1) \ll P(S_{i+1} = SE_2 | S_i = SE_1)$$

$$P(S_{i+1} = SH_2 | S_i = SE_2) \ll P(S_{i+1} = SH_2 | S_i = SH_1)$$

.....

This will result in lowering the user's average logarithm probability value and causing a miscarriage of justice.

Therefore, it needs to establish HMM models for different classified content and monitor them respectively.

Noted that users' access behavior description is based on users' access links and the capture of the links is deployed before the server and firewall avoiding the interferer from services caching and proxy so as to reflect users' true access. User's single visit to different contents of one page will not

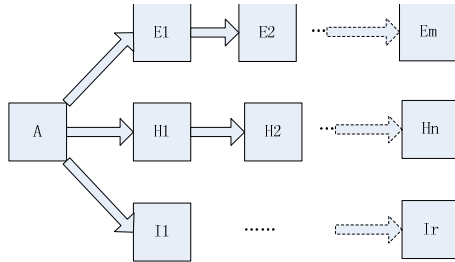


Fig.1 Browsing sequence in classified web pages

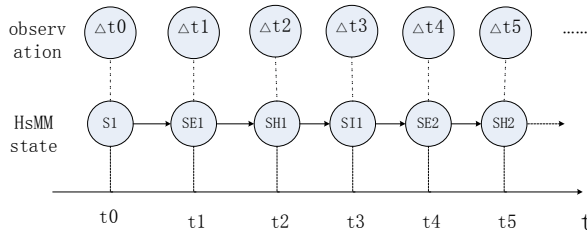


Fig.2 State transition sequence and observation sequence computed repeatedly to simplify the model reasonably.

III. MULTI-CHAINS HMM

A. Modeling

According to the similarity among service contents provided by the web, all links are divided into K categories such as entertainment, sports, IT, real estate, economy, politics, etc.. In fact most websites are organized according to the category of the content. Therefore, this paper adopts the classification of the site architecture.

Assumed that users' navigation behavior to every kind of pages is accordance with Hidden Markov Model and K chains of viewing state are defined to describe user's browsing behavior to K categories of pages. States of each chain provide the same class of service. For one class, different states in the Markov chain represent corresponding clicks on different links. The collection of states are expressed as $S' = \{1, 2, 3, \dots, N'\}$, $l = 1, 2, \dots, K$,

wherein, N' represents the number of states of class l .

For a given state, the time interval between the access requests of adjacent pages which belong to the same class as the given state is an independent random variable of the state and it can be seen as a observation of the given state. In this paper, observations are recorded in seconds and the collection of observations is defined as:

$V' = \{1, 2, \dots, VT\}$, wherein VT is a value big enough and for any normal users, browsing process to any web pages can be completed during VT seconds.

Establish a HMM model λ' for each class l and each model can be denoted by a quintuple (S', V', A', B', π') , wherein

S' is the state collection of l th HMM;

V' is the observation collection of l th HMM;

A' is the state transition possibility matrix of l th HMM, the size of the matrix is $N' \times N'$ and:

$$A' = [a'_{i,j}] \quad (1)$$

$$a'_{i,j} = P(q'_{t+1} = j | q'_t = i), 1 \leq i, j \leq N' \quad (2)$$

B' is the probability distribution of the observations in l th class. $B' = \{b'_j(k)\}$ and $b'_j(k)$ presents the probability of the observation value of state j being k in l th HMM model, i.e.

$$b_j^l(k) = P(k | j), 1 \leq k \leq VT, 1 \leq j \leq N^l \quad (3).$$

π^l is the probability distribution of the initial state in l th class. $\pi^l = \{\pi_i^l\}$ and $\pi_i^l = P(q_1^l = i)$ presents the probability of state in moment 1 being i in l th HMM model.

Due to the complex relationship among links of the website, it is also unavoidable for the user to access pages of different classes alternately. That's to say, there are some state jumps between the multi-chains set up based on classified contents. These state jumps among chains will no longer be considered once the chains have been established.

B. Model application

Considering a user's access behavior, corresponding multiple HMM chains are established when the number of states reaches a predetermined threshold T . Thus corresponding K state transition sequences $q^l = (q_1^l q_2^l q_3^l \cdots q_{Tl}^l)$ and K observation sequences $O^l = (o_1^l o_2^l o_3^l \cdots o_{Tl}^l)$ come into being, in which Tl is length of states of l th Markov chain and $\sum_{l=1}^K Tl = T$. Therefore the probability for the user to browser web pages of l th class is:

$$P(O^l, q^l | \lambda^l) = P(O^l | q^l, \lambda^l) P(q^l | \lambda^l) \quad (4)$$

$$P(O^l | q^l, \lambda^l) = b_{q_1^l}^l(o_1) b_{q_2^l}^l(o_2) b_{q_3^l}^l(o_3) \cdots b_{q_{Tl}^l}^l(o_{Tl}) \quad (5)$$

$$P(q^l | \lambda^l) = \pi_{q_1^l}^l a_{q_1^l q_2^l}^l a_{q_2^l q_3^l}^l \cdots a_{q_{Tl-1}^l q_{Tl}^l}^l \quad (6)$$

For the l th HMM, the mean logarithmic probability of all the sequences of l th class in the training dataset is $mean^l$. The distance between one test chain and $mean^l$ is defined as:

$$bias^l = \left| \frac{\log P(O^l, q^l | \lambda^l) - mean^l}{mean^l} \right| \quad (7)$$

Then the user's overall browsing behavior is normal or not can be defined as followed:

$$Normal = \prod_{l=1}^K u(bias^l - 1) \quad (8)$$

The user's browsing behavior is judged to be normal when $Normal = 1$ and abnormal when $Normal = 0$.

C. Estimating model parameter

When processing the training data in this paper, state transition sequences are extracted besides observation sequences. This enables supervised parameter learning possible where the parameters of K HMMs are estimated using maximum likelihood estimation method.

Assumed that $Count_{i,j}^l$ presents the count of the shift from state i to state j in the training data, then

$$\hat{a}_{i,j}^l = \frac{Count_{i,j}^l}{\sum_{j=1}^{N^l} Count_{i,j}^l} \text{ is the estimation of } a_{i,j}^l;$$

Assumed that for the l th HMM, the count that the observation of state j taking k in the training data is $Count_{j,k}^l$, then

$$\hat{b}_j^l(k) = \frac{Count_{j,k}^l}{\sum_{k=1}^{VT} Count_{j,k}^l} \text{ is the estimation of } b_j^l(k);$$

Assumed that for the l th HMM, the count that the initial state taking i in the training data is $Count_i^l$, then

$$\hat{\pi}_i^l = \frac{Count_i^l}{\sum_{i=1}^{N^l} Count_i^l} \text{ is the estimation of } \pi_i^l;$$

Besides, considering whether the training data is many enough or not, this paper also attempts unsupervised parameter learning and adopts Baum-Welch algorithm to estimate parameters. The comparison between above two methods in parameters estimation is provided in the end.

D. Data preprocessing

1) State discrimination.

For Web applications, one click on a Web page usually contains a request to the master document and many requests to objects embedded in this page (for example: pictures, advertising). According to the definition of state in the paper, request to embedded object is called child state and one state contains all single requests for child state under said state. There are two problems to be settled in switching request sequences received at the server to the state sequences. Firstly, part requests during user access may be answered by the cache of local computer or proxy server, which leads to requests received at the server incomplete. In another word, a state does not necessarily mean that all child states occur. Secondly, users may browse multiple pages at the same time. Hence requests for these web pages may occur alternately, i.e., multiple child states belonging to different states may

appear alternately. In conclusion, whether one state ends up or not cannot be determined based on the occurrence of the child state of another state.

In view of above two aspects, this paper defines that one state reaches its end under two conditions: 1. one of its child states request occur more than one times. 2. After the last child state request, N seconds have passed and none of its child state request occurs.

2) State transition and state observation processing.

State transition is defined as followed: the occurrence of a new state judged according to above discriminant criterion means the ending of the last state.

Due to the diversity and complexity of users' accesses, the duration of each state cannot be counted accurately when multiple states are active. Thus the definition of state observation still inherits the original one. That's to say, the beginning time and ending time of a state correspond to the time in distinguishing states.

IV. EXPERIMENTS AND RESULT ANALYSIS

A. Experimental data

Due to restrictions, the website of one department in the school is taken for example to validate the model. All 235 pages are classified into five categories according to the architecture. 63415 access records from 28409 users during 29 days are sampled and all the data is divided into two data sets:

DataSet1: It contains 25858 user accesses in 20 days.

DataSet2: It contains 2551 user accesses in 9 days.

Above two data sets consist of normal users' accesses and no attack data is included. In order to verify the effectiveness of the model in detecting DDoS attack, simulated attack data DataSet3 is presented by sending normal user requests in higher frequency. In detail, the experiments accelerate the user access requests five times in DataSet2.

B. Experimental results

DataSet1 is used as training set and HMM parameters of the five categories are obtained with maximum likelihood estimation algorithm. Then the logarithm likelihood probabilities of data in DataSet1, DataSet2 and DataSet3 are calculated. Noted that in order to view the results intuitively and under the premise of qualitative analysis, the low logarithm likelihood is set to -100 when it is lower than -100. Two categories of probability distribution histogram are shown in Fig. 3 :

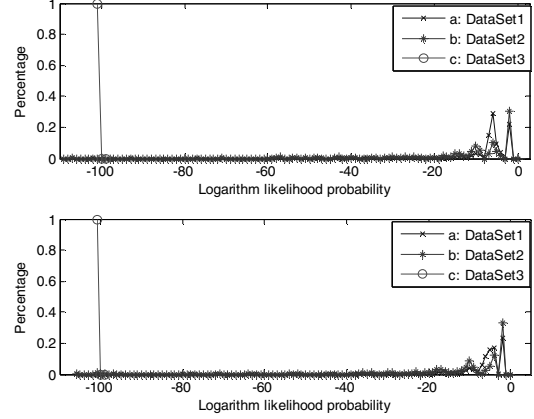


Fig. 3 Logarithm likelihood probability distribution with MSE in multi-chain HMM

Wherein, the upper diagram represents the third class HMM and the lower is the fourth. Curve a represents the logarithm likelihood probabilities histogram distribution of DataSet1, namely the training data. Curve b represents histogram distribution of DataSet2, namely the normal test data. These two curves are roughly the same. Curve c represents the result of DataSet3, i.e. abnormal test data. It can be seen from the graph, the average logarithmic probability of attack data request series is obviously lower than the normal users' access data, which means that this model can effectively difference between normal user access and abnormal attack stream.

Due to the experimental conditions, the experimental links are not many enough which restricts the effectiveness of the maximum likelihood estimation algorithm. Therefore this paper also uses EM algorithm to estimate HMM parameters of five categories. Similarly logarithm likelihood probability distribution histogram of DataSet1 DataSet2 and DataSet3 in the same two categories is shown in Fig.4. Same conclusion can still be concluded from the graph, namely, this model can effectively identify the data.

Additional, HMM with single link is also established when the web pages are not classified. Then the logarithm

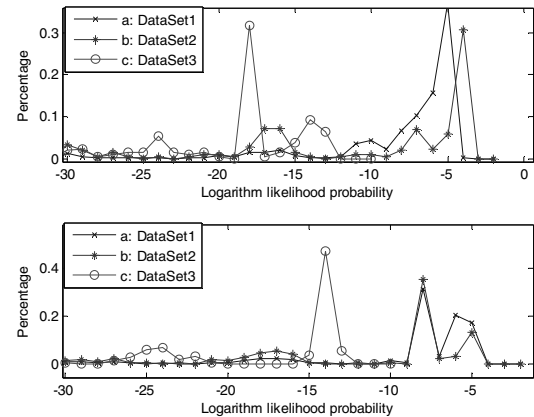


Fig.4 Logarithm likelihood probability distribution with EM in multi-chain HMM

likelihood probability distribution histogram result of DataSet1, DataSet2 and DataSet3 is shown in Fig.5.

As can be seen from the graph, the average logarithm likelihood probabilities of DataSet2 and DataSet3 are significantly lower than DataSet1, which indicates that multi-chains HMM is more efficient than that with single chain in distinguishing normal user access and abnormal ones.

V. CONCLUSION

The paper establishes multiple HMMs for a Web user's access behavior to different category pages in order to further restore and describe user's browse behavior of multi-label browser so as to provide guidance for DDoS intrusion detection. Experimental results show that this method can effectively and feasibly detect abnormalities.

However there still exist some disadvantages in the paper. Firstly, this model judges the category of web pages based on their web site architectures. Actually this classification is not completely accurate and it is more impossible to inoculate users' interests perfectly. Therefore, classifying web pages according to users' access in the training set should be included in the future work. Secondly, treatment against the observations in this model actually magnifies the view time of pages, which means that this model is not able to fully restore users browse behavior. Finally, in the experiment training and testing results are based on a college campus website. The web site is relatively small and most users are school students which means that the composition of users is more onefold. What's more, users' access is convergent to some extent, e.g. many students download the educational form from the website which will influence the model. All of the problems remain to further research and improvement in subsequent work.

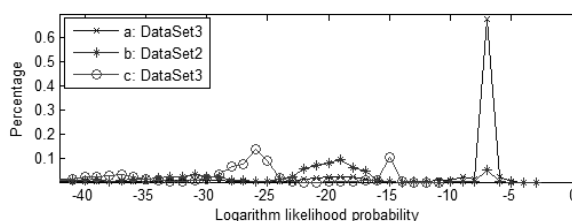


Fig.5 Logarithm likelihood probability distribution with ME in unichain HMM

REFERENCES

- [1] Meyer L, Penzhorn WT. Denial of service and distributed denial of service-today and tomorrow[CA]. In: Proc. Of the IEEE 7th AFRICON Conf[C]. Vol.2,2004,957-964
- [2] REN Xun-yi; WANG Ru-chuan; WANG Hai-yan. Wavelet analysis method for detection of DDoS attack based on self-similar.Journal on Communications[J],Vol.27, No.5, 2006, pp.6-11
- [3] SUN Qin-Dong; ZHANG De-Yun; GAO Peng. Detecting Distributed Denial of Service Attacks Based on Time Series Analysis.[J], Chinese Journal of Computers, Vol.28, No.5, 2005, pp767-773
- [4] Sun Qindong; Zhang Deyun; Sun Zhaohui; Zhang Xiao Detection of Distributed Denial of Service Attacks Based on Flow Connection Density.[J] Journal of Xi'an Jiaotong university. Vol.38, No.10, 2004, pp1048-1052
- [5] GAO Neng FENG Deng-Guo XIANG Ji A Data-Mining Based DoS Detection Technique [J], Chinese Journal of Computers, Vol.29, No.6, 2006, pp944-951
- [6] Tao Peng , Leckie C , Ramamohanarao K. Protection from distributed denial of service attacks using history-based IP filtering [A] . IEEE International Conference on Communications , Anchorage , USA , 2003.
- [7] CERT/ CC Coordination Center . Trends in Denial of Service Attack Technology. October 2001
- [8] XIE Yi , YU Shun-Zheng, A Model for Detecting Application Layer Flooding Attacks,[J], Computer Science, Vol.34, NO.8, 2007, pp109-111
- [9] XIE Yi , YU Shun-Zheng Anomaly Detection Based on Web Users' Browsing Behaviors.[J] Journal of Software, Vol.18, No.4, 2007, pp.967-977